

Computing Integral Length of Sums of Squares and Pythagoras Elements in Real Quadratic Rings

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Let $D \geq 2$ be squarefree and let $\mathcal{O}_D$ be the ring of integers of the real quadratic field $K_D = \mathbb{Q}(\sqrt{D})$. For nonzero $\alpha \in \mathcal{O}_D$ we study the *integral length*

$$\ell_D(\alpha) = \min\{n \in \mathbb{N} : \alpha = x_1^2 + \dots + x_n^2, x_i \in \mathcal{O}_D\},$$

with the convention $\ell_D(\alpha) = \infty$ if no such representation exists. We give an explicit algorithm computing the exact value $\ell_D(\alpha) \in \{1, 2, 3, 4, 5, \infty\}$ for every input element.

The arithmetic input is classical: Peters' criterion decides whether an element of $\mathcal{O}_D$ is a sum of squares at all, and the Pythagoras number of $\mathcal{O}_D$ is known to be at most 5. Our contribution is algorithmic. First, we rewrite Peters' criterion as a nearest-integer test, so representability is decided by a constant number of arithmetic operations in the unit-cost model. Second, we prove that every square occurring in any representation of α lies in an explicit finite search region in the Minkowski embedding. The corrected search bounds are formulated uniformly in terms of $\Delta_D = |\omega_D - \omega'_D|$, which equals $\sqrt{D}$ when $D \equiv 1 \pmod{4}$ and $2\sqrt{D}$ when $D \equiv 2, 3 \pmod{4}$. We then replace the idealised irrational row bounds by an exact integer search: each row is first enclosed by a trace inequality and then recovered by directed binary search using exact sign tests only. This yields a meet-in-the-middle algorithm for the exact length without floating-point arithmetic.

We also revisit the integral Pythagoras number and integral Pythagoras elements. Using the classical classification of Pythagoras numbers of real quadratic orders, together with the exact length algorithm developed here, we obtain an explicit procedure returning both $P(\mathcal{O}_D)$ and an element $a_D \in \mathcal{O}_D$ with $\ell_D(a_D) = P(\mathcal{O}_D)$. Finally, we add a complexity discussion, an exact unit-square balancing step, and a computational study including a balancing ablation on Pell orbits, cross-field exact-length distributions, and a bounded-trace brute-force validation sweep.

CCS Concepts: • **Mathematics of computing** → **Mathematical software**; **Number-theoretic computations**; *Symbolic and algebraic algorithms*; • **Software and its engineering** → *Software libraries and repositories*; • **Theory of computation** → *Design and analysis of algorithms*.

Additional Key Words and Phrases: algorithms, sums of squares, real quadratic fields, rings of integers, exact arithmetic, Pythagoras number, mathematical software

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1 Introduction

Let R be a commutative ring. Determining whether a given element of R is a sum of squares, and if so how many squares are necessary, is one of the oldest representation problems in arithmetic. Over global fields there are now explicit algorithms based on local–global principles and computable local invariants. Darkey-Mensah and Rothkegel computed lengths and Pythagoras elements over arbitrary global fields [3]; Darkey-Mensah adapted the quadratic-form toolkit to global function fields of odd characteristic [2]; Koprowski later gave an explicit algorithm producing a decomposition of minimal length over every global field [8]. Recent companion work treats isotropic vectors over global fields and the computation of dyadic local square-class groups [9, 10]. Outside the global-field line, Kowalczyk and Miska studied Waring numbers of henselian rings [12], and Kowalczyk analysed sums of squares on rational surfaces [11]. In the integral setting, however, local–global principles fail in small rank, and the algorithmic problem changes substantially.

Real quadratic rings form the first natural test case where the arithmetic is both rich and still tractable. Let

$$K_D = \mathbb{Q}(\sqrt{D}), \quad \mathcal{O}_D = \mathcal{O}_{K_D},$$

with $D \geq 2$ squarefree. For $0 \neq \alpha \in \mathcal{O}_D$ we define

$$\ell_D(\alpha) = \min\{n \in \mathbb{N} : \alpha = x_1^2 + \cdots + x_n^2, x_i \in \mathcal{O}_D\},$$

and set $\ell_D(\alpha) = \infty$ if no such representation exists. Our aim is the *element-wise exact-length problem*:

Given D and a specific element $\alpha \in \mathcal{O}_D$, compute the exact value of $\ell_D(\alpha)$.

Several structural results already exist in the literature. Peters characterised those elements of real quadratic orders that are sums of squares and proved an upper bound of 5 for the corresponding Pythagoras number [16, 17]. For maximal real quadratic orders, the exact values of the Pythagoras number are compiled in modern form by Krásenský, Raška and Sgallová [13, Theorem 3.1]; the family-wise problem of deciding when all elements of $m\mathcal{O}_D^+$ are sums of squares was studied by Kala and Yatsyna and then algorithmised by Raška [7, 18]. These papers are structural rather than element-wise algorithmic.

The novelty of the present paper is algorithmic rather than structural. We do not claim new arithmetic input; instead, we isolate a problem that had not previously been written down as an explicit algorithm in the integral real-quadratic setting and solve it in a form suitable both for proof and for exact implementation. More precisely, the present paper contributes:

- (N1) the first explicit *element-wise exact-length algorithm* for maximal real quadratic orders, distinguishing the values 1, 2, 3, 4, 5, ∞ for a given input element of $\mathcal{O}_D$;
- (N2) a constant-cost reformulation of Peters’ representability criterion as a nearest-integer parity test in the unit-cost model;
- (N3) an exact admissible-parallelogram search procedure that can be implemented with integer arithmetic only, via row-wise binary search in the Minkowski embedding rather than floating-point root bounds;
- (N4) a unit-square balancing step that reduces skewed inputs to the norm-sized regime before the finite search begins.

The canonical search algorithm stated in the paper is the same exact trace-enclosure plus sign/bisection routine used in the companion repository; there is no separate floating-point or idealised implementation layer hidden behind the analysis. Compared with the field algorithms in [2, 3, 8–10], the present problem is integral rather than field-valued: the input lies in the order $\mathcal{O}_D$ itself, the target is an integral decomposition, and the method no longer proceeds through a

local–global principle or local square-class computations. Compared with [11, 12], the present paper is not about Waring numbers of henselian local rings or Pythagoras numbers of geometric rings, but about the exact minimal number of squares for a *given element* of a fixed arithmetic order. In this sense the contribution is best viewed as a first explicit element-wise exact-length algorithm for maximal real quadratic orders. The key point is to separate two tasks.

Representability. Since every sum of squares in O_D is a sum of at most five squares, Peters’ five-square criterion decides whether α is representable at all. We rewrite that criterion as a nearest-integer test; this produces a decision procedure using a constant number of arithmetic operations in the unit-cost model.

Exact length. Once representability is known, every summand in any representation of α is bounded in both real embeddings by $\sqrt{\sigma_i(\alpha)}$. This gives a finite search region in the Minkowski embedding. After enumerating the corresponding finite set of relevant squares, the cases $\ell_D(\alpha) = 1, 2, 3, 4$ are decided by set-membership tests, and the remaining representable case is necessarily 5.

The arithmetic ingredients are classical; the new part is their algorithmic packaging. In particular, this paper provides:

- (i) a representability test requiring only a constant number of arithmetic operations in the unit-cost model;
- (ii) an exact finite-search algorithm for $\ell_D(\alpha) \in \{1, 2, 3, 4, 5, \infty\}$;
- (iii) a corrected formulation of the search bounds, using $\Delta_D = |\omega_D - \omega'_D|$ rather than the incomplete expression $\sqrt{D}$;
- (iv) a complexity discussion strengthened by exact unit-square balancing;
- (v) an explicit algorithm computing $P(O_D)$ and an integral Pythagoras element.

The computational section strengthens this algorithmic viewpoint with three reproducible experiments: a balancing ablation along Pell orbits, cross-field length distributions for several representative orders, and a bounded-trace brute-force validation sweep. A reference implementation, regression tests, and the supplementary scripts used for the computational examples are available at https://github.com/tomaszkania/quadratic_sos. The repository mirrors the exact-arithmetic architecture proved in the paper: search regions are enumerated by integer arithmetic only, the public constructor rejects non-squarefree radicands D , and the exact-length interface follows the paper in treating only nonzero inputs. The same balanced search engine is used both by the package and by the table-generation script.

Software availability and reproducibility

The submitted software component contains the installable Python package, regression tests, non-interactive smoke-test drivers, deterministic data-generation scripts, validation scripts, and an executed notebook. The core package has no runtime dependency outside the Python standard library. The referee-facing command `pythonscripts/run_all_checks.py` regenerates quick smoke-test data, runs the regression suite, and performs a bounded brute-force validation sweep. Full reproduction of the deterministic computational data is provided by `pythonscripts/reproduce_tables.py`. Timing values in the balancing ablation are reported only as machine-dependent diagnostics; the structural search counts and exact lengths are deterministic.

The broader context comes from universal quadratic forms and quadratic Waring invariants over totally real fields. The present problem is equivalent to deciding representability of a specific algebraic integer by the diagonal form $\langle 1, \dots, 1 \rangle$ of small rank, and therefore sits naturally inside the recent arithmetic of universal forms and indecomposables [6, 14].

The paper is organised as follows. Section 2 fixes notation and recalls the classical results of Peters and the quadratic Pythagoras classification. In Section 3 we derive the constant-cost representability test. Section 4 constructs the finite

search set and proves correctness of the exact length algorithm. Section 5 discusses explicit search-set bounds, unit-square balancing and reconstruction of minimal decompositions. Section 6 computes $P(O_D)$ and produces a Pythagoras element. Section 7 contains an end-to-end worked example together with a computational study: a local table over O_{10} , a balancing ablation on Pell orbits, cross-field distributions, and a bounded-trace validation sweep.

2 Preliminaries and classical input

Throughout the paper $D \geq 2$ is squarefree and $K_D = \mathbb{Q}(\sqrt{D})$ is a real quadratic field. Its ring of integers is

$$O_D = \mathbb{Z}[\omega_D], \quad \omega_D = \begin{cases} \sqrt{D}, & D \equiv 2, 3 \pmod{4}, \\ \frac{1 + \sqrt{D}}{2}, & D \equiv 1 \pmod{4}. \end{cases}$$

We write ω'_D for the Galois conjugate of ω_D , so

$$\omega'_D = \begin{cases} -\sqrt{D}, & D \equiv 2, 3 \pmod{4}, \\ \frac{1 - \sqrt{D}}{2}, & D \equiv 1 \pmod{4}. \end{cases}$$

The two real embeddings are denoted by

$$\sigma_1(a + b\omega_D) = a + b\omega_D, \quad \sigma_2(a + b\omega_D) = a + b\omega'_D.$$

We also use the quantity

$$\Delta_D := |\omega_D - \omega'_D| = \begin{cases} 2\sqrt{D}, & D \equiv 2, 3 \pmod{4}, \\ \sqrt{D}, & D \equiv 1 \pmod{4}, \end{cases}$$

which is the covolume of the Minkowski lattice $\sigma(O_D) \subset \mathbb{R}^2$. Indeed, in the basis $(1, 1)$ and (ω_D, ω'_D) the absolute value of the determinant is exactly $|\omega_D - \omega'_D| = \Delta_D$.

Definition 2.1. For $0 \neq \alpha \in O_D$, the *integral length* of α is

$$\ell_D(\alpha) = \min\{n \in \mathbb{N} : \alpha = x_1^2 + \cdots + x_n^2, x_i \in O_D\},$$

if such a representation exists, and $\ell_D(\alpha) = \infty$ otherwise.

The *integral Pythagoras number* of O_D is

$$P(O_D) = \sup\{\ell_D(\alpha) : \alpha \in O_D, \ell_D(\alpha) < \infty\}.$$

An *integral Pythagoras element* is an element $a_D \in O_D$ with $\ell_D(a_D) = P(O_D)$.

We use the norm and trace

$$N(a + b\omega_D) = \sigma_1(a + b\omega_D) \sigma_2(a + b\omega_D), \quad \text{Tr}(a + b\omega_D) = \sigma_1(a + b\omega_D) + \sigma_2(a + b\omega_D).$$

More explicitly,

$$N(a + b\omega_D) = \begin{cases} a^2 - Db^2, & D \equiv 2, 3 \pmod{4}, \\ a^2 + ab - \frac{D-1}{4}b^2, & D \equiv 1 \pmod{4}. \end{cases}$$

2.1 Elementary observations

Lemma 2.2. *Let $0 \neq \alpha \in O_D$.*

- (a) *If α is a sum of squares in O_D , then $\sigma_1(\alpha) > 0$ and $\sigma_2(\alpha) > 0$.*
- (b) *If $D \equiv 2, 3 \pmod{4}$ and $\alpha = a + b\sqrt{D}$ is a sum of squares in $O_D = \mathbb{Z}[\sqrt{D}]$, then b is even.*
- (c) *If α' denotes the Galois conjugate of α , then $\ell_D(\alpha') = \ell_D(\alpha)$.*
- (d) *If $\varepsilon \in O_D^\times$ is a unit, then $\ell_D(\varepsilon^2\alpha) = \ell_D(\alpha)$.*

PROOF. If $\alpha = \sum_i x_i^2$, then for each embedding σ_j we have

$$\sigma_j(\alpha) = \sum_i \sigma_j(x_i)^2.$$

Since $\alpha \neq 0$, not all x_i vanish, so both sums are strictly positive. This proves (a).

If $D \equiv 2, 3 \pmod{4}$ and $x = u + v\sqrt{D}$, then

$$x^2 = u^2 + Dv^2 + 2uv\sqrt{D},$$

so every square has an even coefficient at $\sqrt{D}$. This gives (b). Statements (c) and (d) follow by applying conjugation, respectively multiplication by ε , to a representation of α as a sum of squares. $\square$

Remark 2.3. Lemma 2.2(d) provides a useful optimisation. Since $O_D^\times = \{\pm \varepsilon_D^m : m \in \mathbb{Z}\}$ for a fundamental unit $\varepsilon_D > 1$, one may replace α by a suitable unit-square multiple $\varepsilon_D^{2k}\alpha$ before running the finite search. This does not change the answer and can make the two real embeddings comparable; we exploit this in Section 5.

2.2 Two classical theorems

The first input is Peters' characterisation of sums of squares in real quadratic orders. We use the statement in the form quoted by Raška in modern notation [18, Theorem 7]; for the original sources see Peters' two papers [16, 17].

Theorem 2.4 (Peters). *Let $0 \neq \alpha \in O_D$ be totally positive.*

- (i) *Assume $D \equiv 1 \pmod{4}$ and write $\alpha = a + b\omega_D$ with $a, b \in \mathbb{Z}$. Then α is a sum of squares in O_D if and only if there exists $c \in \mathbb{Z}$ with $c \equiv b \pmod{2}$ and*

$$\frac{2a + b - 2\sqrt{N(\alpha)}}{D} \leq c \leq \frac{2a + b + 2\sqrt{N(\alpha)}}{D}.$$

- (ii) *Assume $D \equiv 2, 3 \pmod{4}$ and write $\alpha = a + b\sqrt{D}$ with $a, b \in \mathbb{Z}$. Then α is a sum of squares in O_D if and only if b is even and there exists $c \in \mathbb{Z}$ such that*

$$\frac{a - \sqrt{N(\alpha)}}{2D} \leq c \leq \frac{a + \sqrt{N(\alpha)}}{2D}.$$

The second input is the exact classification of Pythagoras numbers of maximal real quadratic orders. In the form used here it is collected in [13, Theorem 3.1]; the underlying classical sources are Cohn–Pall, Maaß, Peters and Scharlau [1, 15, 17, 19].

Theorem 2.5. *For the maximal order O_D of $K_D = \mathbb{Q}(\sqrt{D})$ one has*

$$P(O_D) = \begin{cases} 3, & D \in \{2, 3, 5\}, \\ 4, & D \in \{6, 7\}, \\ 5, & \text{otherwise.} \end{cases}$$

In particular, every representable element of O_D is a sum of at most five squares.

Corollary 2.6. *For every nonzero $\alpha \in O_D$,*

$$\ell_D(\alpha) \in \{1, 2, 3, 4, 5, \infty\},$$

and α is a sum of squares if and only if it is a sum of five squares.

PROOF. This is immediate from Theorem 2.5. □

3 Representability by integral squares

Peters' criterion already decides representability. For computation it is convenient to turn the interval condition into a nearest-integer test and to state carefully what is meant by “constant-cost”.

Proposition 3.1. *Let $0 \neq \alpha \in O_D$.*

- (i) *Assume $D \equiv 1 \pmod{4}$ and write $\alpha = a + b\omega_D$. Set $C = 2a + b$ and let*

$$\mathcal{A}_b = \{c \in \mathbb{Z} : c \equiv b \pmod{2}\}.$$

Choose $c_0 \in \mathcal{A}_b$ minimising $|Dc - C|$. Then α is a sum of squares in O_D if and only if

$$\sigma_1(\alpha) > 0, \quad \sigma_2(\alpha) > 0, \quad \text{and} \quad (Dc_0 - C)^2 \leq 4N(\alpha).$$

- (ii) *Assume $D \equiv 2, 3 \pmod{4}$ and write $\alpha = a + b\sqrt{D}$. Choose $c_0 \in \mathbb{Z}$ minimising $|2Dc - a|$. Then α is a sum of squares in O_D if and only if*

$$\sigma_1(\alpha) > 0, \quad \sigma_2(\alpha) > 0, \quad b \equiv 0 \pmod{2}, \quad \text{and} \quad (2Dc_0 - a)^2 \leq N(\alpha).$$

Moreover, the test uses a constant number of arithmetic operations and comparisons in the unit-cost model.

PROOF. By Corollary 2.6, α is representable if and only if it is a sum of five squares. If α is representable, then Lemma 2.2(a) gives total positivity, and in the second case Lemma 2.2(b) gives the parity condition.

Assume first that $D \equiv 1 \pmod{4}$. By Theorem 2.4, representability is equivalent to the existence of an integer $c \equiv b \pmod{2}$ satisfying

$$|Dc - C| \leq 2\sqrt{N(\alpha)}.$$

Among all such integers, the left-hand side is minimised at a nearest admissible integer c_0 . Hence such a c exists if and only if the minimum itself satisfies the same inequality, which is exactly

$$(Dc_0 - C)^2 \leq 4N(\alpha).$$

The proof in the case $D \equiv 2, 3 \pmod{4}$ is identical. By Theorem 2.4, representability is equivalent to b even and the existence of an integer c with

$$|2Dc - a| \leq \sqrt{N(\alpha)}.$$

The minimum is attained at a nearest integer c_0 to $a/(2D)$, and squaring gives the desired criterion. The final assertion is immediate: the procedure uses only a constant number of arithmetic operations on the coefficients of α and D , together with exact comparisons. $\square$

Remark 3.2. Proposition 3.1 is “constant-cost” only in the unit-cost arithmetic model. In the bit model the input size still matters, of course: one performs a constant number of integer additions, multiplications, congruence tests and comparisons on integers whose bit size is $O(\log |D| + \log H)$, where H bounds the coefficients of α .

Algorithm 1: Representability in $\mathcal{O}_D$

Input : A squarefree integer $D \geq 2$ and a nonzero element $\alpha \in \mathcal{O}_D$.

Output: TRUE if α is a sum of squares in $\mathcal{O}_D$, otherwise FALSE.

```

1 if  $\sigma_1(\alpha) \leq 0$  or  $\sigma_2(\alpha) \leq 0$  then
2   return FALSE;
3 if  $D \equiv 1 \pmod{4}$  then
4   Write  $\alpha = a + b\omega_D$  with  $a, b \in \mathbb{Z}$  and set  $C \leftarrow 2a + b$ ;
5   Let  $c_0$  be a nearest integer to  $C/D$  with  $c_0 \equiv b \pmod{2}$ ;
6   if  $(Dc_0 - C)^2 \leq 4N(\alpha)$  then
7     return TRUE;
8   return FALSE;
9 else
10  Write  $\alpha = a + b\sqrt{D}$  with  $a, b \in \mathbb{Z}$ ;
11  if  $b$  is odd then
12    return FALSE;
13  Let  $c_0$  be a nearest integer to  $a/(2D)$ ;
14  if  $(2Dc_0 - a)^2 \leq N(\alpha)$  then
15    return TRUE;
16  return FALSE;
```

Theorem 3.3. *Algorithm 1 is correct.*

PROOF. This is Proposition 3.1 written in pseudocode. $\square$

4 Finite search and exact integral length

We now assume that $\alpha \in \mathcal{O}_D$ is totally positive and construct an explicit finite search set containing every possible summand of every representation of α .

4.1 Corrected embedding bounds

Set

$$A_1 = \sqrt{\sigma_1(\alpha)}, \quad A_2 = \sqrt{\sigma_2(\alpha)}.$$

For $x \in K_D$ we write $x^2 \preccurlyeq \alpha$ if $\sigma_i(x^2) \leq \sigma_i(\alpha)$ for $i = 1, 2$.

Lemma 4.1. *Let $x = u + v\omega_D \in \mathcal{O}_D$ and assume $x^2 \preccurlyeq \alpha$. Then*

$$|v| \leq \frac{A_1 + A_2}{\Delta_D}$$

and

$$|u| \leq \frac{|\omega_D|A_2 + |\omega'_D|A_1}{\Delta_D}.$$

In particular, only finitely many elements $x \in \mathcal{O}_D$ satisfy $x^2 \preccurlyeq \alpha$.

PROOF. From $x^2 \preccurlyeq \alpha$ we obtain

$$|\sigma_1(x)| \leq A_1, \quad |\sigma_2(x)| \leq A_2.$$

Since

$$\sigma_1(x) - \sigma_2(x) = v(\omega_D - \omega'_D),$$

we get

$$|v| = \left| \frac{\sigma_1(x) - \sigma_2(x)}{\omega_D - \omega'_D} \right| \leq \frac{|\sigma_1(x)| + |\sigma_2(x)|}{|\omega_D - \omega'_D|} \leq \frac{A_1 + A_2}{\Delta_D}.$$

This is the point at which the distinction between $D \equiv 1 \pmod{4}$ and $D \equiv 2, 3 \pmod{4}$ matters: one has $\Delta_D = \sqrt{D}$ in the former case and $\Delta_D = 2\sqrt{D}$ in the latter.

Next, solving the linear system

$$\sigma_1(x) = u + v\omega_D, \quad \sigma_2(x) = u + v\omega'_D$$

for u yields

$$u = \frac{\omega_D \sigma_2(x) - \omega'_D \sigma_1(x)}{\omega_D - \omega'_D}.$$

Therefore

$$|u| \leq \frac{|\omega_D| |\sigma_2(x)| + |\omega'_D| |\sigma_1(x)|}{|\omega_D - \omega'_D|} \leq \frac{|\omega_D|A_2 + |\omega'_D|A_1}{\Delta_D}.$$

Since $u, v \in \mathbb{Z}$, only finitely many pairs (u, v) are possible. □

Corollary 4.2. *Define*

$$U_\alpha := \left\lfloor \frac{|\omega_D|A_2 + |\omega'_D|A_1}{\Delta_D} \right\rfloor, \quad V_\alpha := \left\lfloor \frac{A_1 + A_2}{\Delta_D} \right\rfloor.$$

Then every $x = u + v\omega_D \in \mathcal{O}_D$ with $x^2 \preccurlyeq \alpha$ satisfies $|u| \leq U_\alpha$ and $|v| \leq V_\alpha$.

Definition 4.3. For totally positive $\alpha \in \mathcal{O}_D$ define the finite search set

$$B(\alpha) = \{x \in \mathcal{O}_D : x^2 \preccurlyeq \alpha\}.$$

Let

$$S(\alpha) = \{x^2 : x \in B(\alpha)\} \subseteq \mathcal{O}_D$$

be the corresponding set of relevant squares, and define

$$T_2(\alpha) = \{s_1 + s_2 : s_1, s_2 \in S(\alpha)\}.$$

By Corollary 4.2, the set $B(\alpha)$ is finite.

Proposition 4.4 (Geometric row description). *For each integer v , define*

$$u_{\min}(v) = \left\lceil \max\{-A_1 - v\omega_D, -A_2 - v\omega'_D\} \right\rceil$$

and

$$u_{\max}(v) = \lfloor \min\{A_1 - v\omega_D, A_2 - v\omega'_D\} \rfloor.$$

Then

$$\{u \in \mathbb{Z} : u + v\omega_D \in B(\alpha)\} = \{u \in \mathbb{Z} : u_{\min}(v) \leq u \leq u_{\max}(v)\}.$$

In particular, each fixed- v row of $B(\alpha)$ is an interval of integers or is empty.

PROOF. The inequalities $|\sigma_1(u + v\omega_D)| \leq A_1$ and $|\sigma_2(u + v\omega_D)| \leq A_2$ are equivalent to

$$-A_1 - v\omega_D \leq u \leq A_1 - v\omega_D$$

and

$$-A_2 - v\omega'_D \leq u \leq A_2 - v\omega'_D.$$

Hence an integer u is admissible if and only if it lies in the intersection of these two intervals, which is exactly the range $u_{\min}(v) \leq u \leq u_{\max}(v)$. $\square$

Proposition 4.5 (Exact trace enclosure for a fixed row). *Let $\alpha \in \mathcal{O}_D$ be totally positive and fix $v \in \mathbb{Z}$.*

(i) *If $D \equiv 2, 3 \pmod{4}$ and $x = u + v\sqrt{D} \in B(\alpha)$, then*

$$u^2 \leq \frac{\text{Tr}(\alpha)}{2} - Dv^2.$$

Hence the admissible integers u lie in the interval

$$I_v^{\text{tr}}(\alpha) = \left[- \left\lfloor \sqrt{\frac{\text{Tr}(\alpha)}{2} - Dv^2} \right\rfloor, \left\lfloor \sqrt{\frac{\text{Tr}(\alpha)}{2} - Dv^2} \right\rfloor \right]$$

whenever $\text{Tr}(\alpha)/2 - Dv^2 \geq 0$, and the row is empty otherwise.

(ii) *If $D \equiv 1 \pmod{4}$ and $x = u + v\omega_D \in B(\alpha)$, then*

$$(2u + v)^2 \leq 2\text{Tr}(\alpha) - Dv^2.$$

Hence the admissible integers u lie in the interval

$$I_v^{\text{tr}}(\alpha) = \left[\left\lfloor \frac{-v - \lfloor \sqrt{2\text{Tr}(\alpha) - Dv^2} \rfloor}{2} \right\rfloor, \left\lfloor \frac{-v + \lfloor \sqrt{2\text{Tr}(\alpha) - Dv^2} \rfloor}{2} \right\rfloor \right]$$

whenever $2\text{Tr}(\alpha) - Dv^2 \geq 0$, and the row is empty otherwise.

In particular, rows can occur only for

$$|v| \leq V_{\alpha}^{\text{tr}} := \begin{cases} \left\lfloor \sqrt{\frac{2\text{Tr}(\alpha)}{D}} \right\rfloor, & D \equiv 1 \pmod{4}, \\ \left\lfloor \sqrt{\frac{\text{Tr}(\alpha)}{2D}} \right\rfloor, & D \equiv 2, 3 \pmod{4}. \end{cases}$$

PROOF. Write $x = u + v\omega_D$ or $x = u + v\sqrt{D}$ according to the congruence class of D modulo 4. Since $x^2 \preccurlyeq \alpha$, one has $\text{Tr}(x^2) \leq \text{Tr}(\alpha)$. If $D \equiv 2, 3 \pmod{4}$, then

$$\text{Tr}((u + v\sqrt{D})^2) = 2u^2 + 2Dv^2,$$

so $2u^2 + 2Dv^2 \leq \text{Tr}(\alpha)$, which is exactly (i). If $D \equiv 1 \pmod{4}$, then using $\omega_D^2 = (D-1)/4 + \omega_D$ and $\text{Tr}(a + b\omega_D) = 2a + b$,

$$\text{Tr}((u + v\omega_D)^2) = 2u^2 + 2uv + \frac{D+1}{2}v^2.$$

Multiplying by 2 and completing the square gives

$$(2u + v)^2 + Dv^2 \leq 2 \text{Tr}(\alpha),$$

which is equivalent to (ii). The bound on $|v|$ is immediate in both cases. $\square$

Proposition 4.6 (Exact row search by directed bisection). *Let $\alpha \in \mathcal{O}_D$ be totally positive and fix $v \in \mathbb{Z}$. Set*

$$J_v(\alpha) = \{u \in I_v^{\text{tr}}(\alpha) : u + v\omega_D \in B(\alpha)\}.$$

Then $J_v(\alpha)$ is an interval of integers or is empty. Moreover:

- (a) *one can determine whether $J_v(\alpha)$ is empty and, if not, find one admissible integer $u_0 \in J_v(\alpha)$ in $O(\log(1 + |I_v^{\text{tr}}(\alpha)|))$ exact arithmetic operations;*
- (b) *once such a u_0 is known, the left and right endpoints of $J_v(\alpha)$ are found by two further binary searches of the same complexity;*
- (c) *consequently, $B(\alpha)$ and $S(\alpha)$ can be enumerated exactly, row by row, without any floating-point arithmetic.*

PROOF. By Proposition 4.4, the admissible integers on a fixed row already form an interval or are empty, so only the exact search remains to justify. For a trial value $u \in I_v^{\text{tr}}(\alpha)$ put $x = u + v\omega_D$. The admissibility test $x \in B(\alpha)$ is equivalent to

$$\sigma_1(\alpha - x^2) \geq 0 \quad \text{and} \quad \sigma_2(\alpha - x^2) \geq 0,$$

which is decidable by exact sign tests on elements of $\mathcal{O}_D$. Assume that $x \notin B(\alpha)$. For each violated embedding $i \in \{1, 2\}$ one has $|\sigma_i(x)| > A_i$. Because the map $u \mapsto \sigma_i(u + v\omega_D)$ is affine with slope 1, the sign of $\sigma_i(x)$ determines on which side of the admissible interval for the i -th embedding the point lies: if $\sigma_i(x) < 0$, then every admissible integer lies strictly to the right of u ; if $\sigma_i(x) > 0$, then every admissible integer lies strictly to the left. These signs are again computable exactly. If the two violated embeddings prescribe opposite directions, then the two embedding intervals are disjoint and $J_v(\alpha) = \emptyset$; otherwise the common prescribed half-interval contains all admissible points. Thus a directed binary search halves the current search interval at each step until either an admissible point is found or emptiness is certified. This proves (a).

Once one admissible point u_0 is found, the interval property of $J_v(\alpha)$ allows one to locate its left and right endpoints by ordinary binary search using only the admissibility test. This proves (b), and (c) follows by scanning all integers between the two exact endpoints on every nonempty row. $\square$

Remark 4.7. The geometric description in Proposition 4.4 is the two-dimensional analogue of enumerating lattice points inside the actual search body rather than its axis-aligned bounding box. In higher-dimensional lattice algorithms this viewpoint is standard in Fincke–Pohst style enumeration [5]; the point of Proposition 4.6 is that in the present dimension the same idea admits a completely exact implementation by trace enclosures and integer binary search.

Lemma 4.8 (Completeness of the search set). *Let $\alpha \in \mathcal{O}_D$ be totally positive and suppose*

$$\alpha = x_1^2 + \cdots + x_n^2 \quad (x_i \in \mathcal{O}_D).$$

Then each x_i belongs to $B(\alpha)$, and hence each square x_i^2 belongs to $S(\alpha)$.

PROOF. For each embedding σ_j and each index i one has

$$\sigma_j(\alpha) = \sum_{k=1}^n \sigma_j(x_k)^2 \geq \sigma_j(x_i)^2.$$

Therefore $|\sigma_j(x_i)| \leq A_j$ for $j = 1, 2$, so Lemma 4.1 applies and shows that $x_i \in B(\alpha)$. $\square$

4.2 Exact-length criterion

Theorem 4.9. *Let $0 \neq \alpha \in \mathcal{O}_D$.*

- (i) $\ell_D(\alpha) = 1$ if and only if $\alpha \in S(\alpha)$.
- (ii) $\ell_D(\alpha) = 2$ if and only if $\alpha \notin S(\alpha)$ and $\alpha \in T_2(\alpha)$.
- (iii) $\ell_D(\alpha) = 3$ if and only if $\alpha \notin S(\alpha) \cup T_2(\alpha)$ and there exists $s \in S(\alpha)$ with $\alpha - s \in T_2(\alpha)$.
- (iv) $\ell_D(\alpha) = 4$ if and only if none of the previous cases occurs and there exists $t \in T_2(\alpha)$ with $\alpha - t \in T_2(\alpha)$.
- (v) If Algorithm 1 returns TRUE and none of (i)–(iv) holds, then $\ell_D(\alpha) = 5$.
- (vi) If Algorithm 1 returns FALSE, then $\ell_D(\alpha) = \infty$.

PROOF. If Algorithm 1 returns FALSE, then by Theorem 3.3 the element α is not a sum of squares in $\mathcal{O}_D$, so $\ell_D(\alpha) = \infty$. This proves (vi).

Assume from now on that Algorithm 1 returns TRUE. By Corollary 2.6, the remaining possibilities are 1, 2, 3, 4, 5.

Assertion (i) is immediate from the definition of $S(\alpha)$.

For (ii), if $\alpha = s_1 + s_2$ is a sum of two squares, then Lemma 4.8 gives $s_1, s_2 \in S(\alpha)$, hence $\alpha \in T_2(\alpha)$. Conversely, any identity $\alpha = s_1 + s_2$ with $s_1, s_2 \in S(\alpha)$ is by definition a representation by two squares. Excluding the case $\alpha \in S(\alpha)$ forces the minimal length to be exactly 2.

For (iii), a representation by three squares has the form $\alpha = s_1 + s_2 + s_3$ with $s_i \in S(\alpha)$ by Lemma 4.8. Equivalently, $\alpha - s_1 = s_2 + s_3 \in T_2(\alpha)$ for some $s_1 \in S(\alpha)$. Conversely, any such identity yields a representation by three squares. Excluding lengths 1 and 2 forces the minimal length to be exactly 3.

For (iv), if $\alpha = s_1 + s_2 + s_3 + s_4$ with each $s_i \in S(\alpha)$, then $\alpha = (s_1 + s_2) + (s_3 + s_4) = t_1 + t_2$ with $t_1, t_2 \in T_2(\alpha)$. Conversely, if $\alpha = t_1 + t_2$ with $t_1, t_2 \in T_2(\alpha)$, then by definition there exist $s_1, s_2, s_3, s_4 \in S(\alpha)$ such that $t_1 = s_1 + s_2$ and $t_2 = s_3 + s_4$, so $\alpha = s_1 + s_2 + s_3 + s_4$ is a representation by four squares. Excluding the previous cases forces the minimal length to be exactly 4.

Finally, if none of (i)–(iv) holds but α is representable, then the only remaining possibility allowed by Corollary 2.6 is $\ell_D(\alpha) = 5$. This proves (v). $\square$

Theorem 4.10. *Algorithm 2 (below) is correct. Its core search uses only exact integer arithmetic and exact sign tests in $\mathcal{O}_D$; in particular, no floating-point approximation is required.*

PROOF. Proposition 4.5 supplies a finite family of rows containing all possible elements of $B(\alpha)$. On each row, Proposition 4.6 recovers the exact admissible interval of integers u , so the algorithm inserts precisely the squares x^2 with $x \in B(\alpha)$. The final assertion is part (c) of Proposition 4.6. $\square$

Algorithm 2: Enumerating the relevant squares by exact row search**Input :** A squarefree integer $D \geq 2$ and a totally positive element $\alpha \in \mathcal{O}_D$.**Output :** The finite set $S(\alpha)$ of all squares x^2 with $x^2 \preccurlyeq \alpha$.

```

1 Compute the trace row bound  $V_\alpha^{\text{tr}}$  from Proposition 4.5;
2 Initialise  $S \leftarrow \emptyset$ ;
3 for  $v = -V_\alpha^{\text{tr}}$  to  $V_\alpha^{\text{tr}}$  do
4   Compute the trace enclosure  $I_v^{\text{tr}}(\alpha)$  from Proposition 4.5;
5   if  $I_v^{\text{tr}}(\alpha)$  is empty then
6      $\quad$  continue;
7   Use Proposition 4.6 to determine the exact admissible row
      
$$J_v(\alpha) = \{u \in I_v^{\text{tr}}(\alpha) : (u + v\omega_D)^2 \preccurlyeq \alpha\}.$$

      if  $J_v(\alpha)$  is empty then
8      $\quad$  continue;
9   Let  $u_{\min}(v) \leq u_{\max}(v)$  be the two endpoints of  $J_v(\alpha)$ ;
10  for  $u = u_{\min}(v)$  to  $u_{\max}(v)$  do
11     $\quad$  Insert  $(u + v\omega_D)^2$  into  $S$ ;
12 return  $S$ ;
```

Theorem 4.11. *Algorithm 3 (below) is correct.***PROOF.** Algorithm 1 decides representability by Theorem 3.3. If it answers negatively, the output ∞ is correct.

Assume from now on that α is representable, and let $\beta = u^{2k}\alpha$ be the balanced unit-square multiple computed in the second line. By Lemma 2.2(d), $\ell_D(\beta) = \ell_D(\alpha)$. By Theorem 4.10, Algorithm 2 returns the exact set $S(\beta)$. The test $\beta \in S(\beta)$ is correct by Theorem 4.9(i). Next, β has length 2 if and only if $\beta = s_1 + s_2$ for some $s_1, s_2 \in S(\beta)$; equivalently, there exists $s \in S(\beta)$ with $\beta - s \in S(\beta)$. Thus the second loop decides the case $\ell_D(\beta) = 2$ and hence $\ell_D(\alpha) = 2$.

If $p = P(\mathcal{O}_D) = 3$, then every representable element has length at most 3 by Theorem 2.5, so after excluding lengths 1 and 2 the algorithm may correctly return 3. Assume therefore that $p \geq 4$. The next loop checks whether $\beta - s \in T_2(\beta)$ for some $s \in S(\beta)$, which is equivalent to $\ell_D(\beta) = 3$ by Theorem 4.9(iii) and therefore to $\ell_D(\alpha) = 3$.

If $p = 4$, then every representable element has length at most 4, so after excluding lengths 1, 2, 3 the algorithm may correctly return 4. Finally, when $p = 5$, the last loop is exactly the four-square test from Theorem 4.9(iv) applied to β , and failure of that test forces $\ell_D(\beta) = 5$ by Theorem 4.9(v). Since $\ell_D(\beta) = \ell_D(\alpha)$, the returned value is correct. $\square$

Algorithm 3: Exact integral length in O_D **Input** : A squarefree integer $D \geq 2$ and a nonzero element $\alpha \in O_D$.**Output**: The exact value of $\ell_D(\alpha)$ in $\{1, 2, 3, 4, 5, \infty\}$.

```

1 if Algorithm 1 returns FALSE then
2   return  $\infty$ ;
3 Compute a balanced unit-square multiple  $\beta = u^{2k}\alpha$  as in Lemma 5.4;
4 Determine  $p \leftarrow P(O_D)$  from Theorem 2.5;
5 Compute  $S \leftarrow S(\beta)$  using Algorithm 2 and store it as a hash set;
6 if  $\beta \in S$  then
7   return 1;
8 foreach  $s \in S$  do
9   if  $\beta - s \in S$  then
10    return 2;
11 if  $p = 3$  then
12   return 3;
13 Construct the pair-sum set
      
$$T_2 \leftarrow \{s_1 + s_2 : s_1, s_2 \in S\}$$

      and store it as a hash set;
14 foreach  $s \in S$  do
15   if  $\beta - s \in T_2$  then
16     return 3;
17 if  $p = 4$  then
18   return 4;
19 foreach  $t \in T_2$  do
20   if  $\beta - t \in T_2$  then
21     return 4;
22 return 5;

```

Remark 4.12 (Ternary short-circuits and Maaß’s criterion). The short-circuit at $p = 3$ removes the construction of $T_2(\beta)$ in the cases $D \in \{2, 3, 5\}$. In particular, for $D = 5$ one could also replace the uniform three-square step by Maaß’s specialised ternary criterion [15]. We keep the present uniform algorithm because it applies unchanged to every maximal real quadratic order.

5 Complexity, balancing, and constructive output

We first state the cost of the search-and-decision phase after balancing, then derive explicit bounds for the size of the resulting finite square set, and finally explain how to recover an actual minimal decomposition.

Theorem 5.1. *Let $\alpha \in \mathcal{O}_D$ be representable and let $\beta = u^{2k}\alpha$ be the balanced unit-square multiple supplied to the search-and-decision phase of Algorithm 3. Put*

$$M(\beta) = |S(\beta)|, \quad N_B(\beta) = |B(\beta)|, \quad \text{and} \quad \Sigma_{\text{row}}(\beta) = \sum_{|v| \leq V_\beta^{\text{tr}}} \log(1 + |I_v^{\text{tr}}(\beta)|),$$

where V_β^{tr} and the trace row intervals $I_v^{\text{tr}}(\beta)$ are given by Proposition 4.5. Then the search-and-decision phase of Algorithm 3 after balancing can be implemented as follows.

- (a) By Algorithm 2, one enumerates $B(\beta)$ and builds the hash set $S(\beta)$ in

$$O(N_B(\beta) + \Sigma_{\text{row}}(\beta))$$

exact arithmetic operations and $O(M(\beta))$ memory.

- (b) The tests for lengths 1 and 2 require one hash lookup, respectively $M(\beta)$ hash lookups in $S(\beta)$, so they take expected time $O(1)$ and $O(M(\beta))$ after $S(\beta)$ has been built.
- (c) If $P(\mathcal{O}_D) \geq 4$, one constructs $T_2(\beta)$ in expected time $O(M(\beta)^2)$ and memory $O(M(\beta)^2)$. The subsequent three-square test uses only $M(\beta)$ additional hash lookups in $T_2(\beta)$.
- (d) If $P(\mathcal{O}_D) = 5$, the four-square test scans $T_2(\beta)$ and performs $|T_2(\beta)| \leq M(\beta)^2$ further hash lookups in $T_2(\beta)$.

Consequently, after the balanced representative β has been computed, the remaining search-and-decision phase uses

$$\text{time} = O(N_B(\beta) + \Sigma_{\text{row}}(\beta) + M(\beta)^2), \quad \text{memory} = O(M(\beta)^2)$$

in expected time with hash tables. In the ternary cases $D \in \{2, 3, 5\}$, the pair-sum set $T_2(\beta)$ need not be constructed, so the expected running time improves to

$$O(N_B(\beta) + \Sigma_{\text{row}}(\beta) + M(\beta)).$$

The representability test of Algorithm 1 contributes only a constant number of arithmetic operations in the unit-cost model. In the present implementation, the balancing step itself is carried out iteratively and contributes an additional $O(|k|)$ exact unit multiplications and ratio comparisons, where $\beta = u^{2k}\alpha$.

PROOF. Algorithm 2 scans exactly the $N_B(\beta)$ admissible roots and, by Proposition 4.6, performs on each row one directed bisection together with two boundary bisections. This gives the cost $O(N_B(\beta) + \Sigma_{\text{row}}(\beta))$ for building $S(\beta)$. The tests for lengths 1 and 2 use only membership in the hash set $S(\beta)$, so their costs are immediate. Once $T_2(\beta)$ is built, the three-square test scans $S(\beta)$ and checks whether $\beta - s \in T_2(\beta)$ for each $s \in S(\beta)$, while the four-square test scans $T_2(\beta)$ and checks whether $\beta - t \in T_2(\beta)$ for each $t \in T_2(\beta)$. This yields the stated time and memory bounds. In the ternary cases the short-circuit in Algorithm 3 avoids the construction of $T_2(\beta)$ altogether. $\square$

Remark 5.2 (Space reduction by sorted pair-sum streams). The explicit construction of $T_2(\beta)$ is convenient but uses quadratic memory. Since the four-square step is a collision problem between two pair-sum streams, one may instead adapt the classical Schroeppe–Shamir meet-in-the-middle technique [20] to lower the memory requirement to $O(M(\beta))$, at the cost of sorted-stream bookkeeping and an extra logarithmic factor in time. We retain the simpler explicit construction of $T_2(\beta)$ in the main algorithm.

Proposition 5.3. *For every totally positive $\alpha \in \mathcal{O}_D$ one has*

$$M(\alpha) \leq |B(\alpha)| \leq (2U_\alpha + 1)(2V_\alpha + 1),$$

where U_α and V_α are given by Corollary 4.2. In particular,

$$M(\alpha) = O(U_\alpha V_\alpha).$$

PROOF. The inequality $M(\alpha) \leq |B(\alpha)|$ is immediate from the definition of $S(\alpha)$. The second inequality follows because there are at most $2U_\alpha + 1$ admissible integers u and at most $2V_\alpha + 1$ admissible integers v . $\square$

The bound in Proposition 5.3 is fully explicit, but it depends on the imbalance of the two embeddings. For a fixed real quadratic ring one can sharpen the complexity statement by using the unit-square invariance from Lemma 2.2(d).

Lemma 5.4. *Fix a fundamental unit $\varepsilon_D > 1$ of O_D . For every totally positive $\alpha \in O_D$ there exists $k \in \mathbb{Z}$ such that*

$$\beta = \varepsilon_D^{2k} \alpha$$

satisfies

$$\varepsilon_D^{-2} \leq \frac{\sigma_1(\beta)}{\sigma_2(\beta)} < \varepsilon_D^2.$$

Moreover, $\ell_D(\beta) = \ell_D(\alpha)$ and $N(\beta) = N(\alpha)$.

PROOF. Set

$$r = \frac{\sigma_1(\alpha)}{\sigma_2(\alpha)} > 0.$$

Since $\sigma_1(\varepsilon_D^2) = \varepsilon_D^2$ and $\sigma_2(\varepsilon_D^2) = \varepsilon_D^{-2}$, multiplying by ε_D^{2k} changes the ratio by a factor of ε_D^{4k} . Choose $k \in \mathbb{Z}$ so that

$$\varepsilon_D^{-2} \leq r \varepsilon_D^{4k} < \varepsilon_D^2.$$

This is possible because the intervals $[\varepsilon_D^{4m-2}, \varepsilon_D^{4m+2})$, $m \in \mathbb{Z}$, cover $(0, \infty)$. The equality of lengths is Lemma 2.2(d), and $N(\beta) = N(\varepsilon_D)^{2k} N(\alpha) = N(\alpha)$ because $N(\varepsilon_D) = \pm 1$. $\square$

Remark 5.5. For the complexity statements it is natural to use a fundamental unit, since this gives the smallest balancing constant. For the implementation, however, any fixed totally positive unit $u > 1$ suffices: replacing ε_D by u leaves the length invariant and yields the same balancing mechanism, with constants depending on u . The companion repository uses a positive norm-one Pell unit obtained from the continued fraction of $\sqrt{D}$, which keeps the balancing step exact while avoiding floating-point logarithms.

Corollary 5.6. *Let $\beta = \varepsilon_D^{2k} \alpha$ be balanced as in Lemma 5.4. Then*

$$A_1(\beta), A_2(\beta) \leq \varepsilon_D^{1/2} N(\alpha)^{1/4},$$

and consequently

$$U_\beta \leq \varepsilon_D^{1/2} N(\alpha)^{1/4}, \quad V_\beta \leq \frac{2\varepsilon_D^{1/2}}{\Delta_D} N(\alpha)^{1/4}.$$

In particular,

$$M(\beta) = O_D \left(\frac{\sqrt{N(\alpha)}}{\Delta_D} \right),$$

and, once the balanced representative β has been computed, the search-and-decision phase of Algorithm 3 runs in expected time

$$O_D \left(\frac{N(\alpha)}{\Delta_D^2} \right) = O_D \left(\frac{N(\alpha)}{D} \right)$$

in the unit-cost model. The current implementation reaches β by iterative Pell-unit balancing and therefore adds $O(|k|)$ exact preprocessing steps, where $\beta = \varepsilon_D^{2k} \alpha$.

PROOF. Put $r = \sigma_1(\beta)/\sigma_2(\beta)$. By Lemma 5.4, both r and r^{-1} are at most ε_D^2 . Since β is totally positive, $N(\beta) = N(\alpha) > 0$, and therefore

$$\sigma_1(\beta)^2 = N(\alpha) r, \quad \sigma_2(\beta)^2 = N(\alpha) r^{-1}.$$

Taking positive square roots gives

$$\sigma_1(\beta) = N(\alpha)^{1/2} r^{1/2}, \quad \sigma_2(\beta) = N(\alpha)^{1/2} r^{-1/2},$$

whence

$$A_1(\beta) = \sigma_1(\beta)^{1/2} \leq \varepsilon_D^{1/2} N(\alpha)^{1/4}, \quad A_2(\beta) = \sigma_2(\beta)^{1/2} \leq \varepsilon_D^{1/2} N(\alpha)^{1/4}.$$

Now $|\omega_D| + |\omega'_D| = \Delta_D$ in both congruence classes modulo 4, so Corollary 4.2 gives

$$U_\beta \leq \frac{|\omega_D| A_2(\beta) + |\omega'_D| A_1(\beta)}{\Delta_D} \leq \varepsilon_D^{1/2} N(\alpha)^{1/4}$$

and

$$V_\beta \leq \frac{A_1(\beta) + A_2(\beta)}{\Delta_D} \leq \frac{2\varepsilon_D^{1/2}}{\Delta_D} N(\alpha)^{1/4}.$$

The estimate for $M(\beta)$ follows from Proposition 5.3. Moreover,

$$\sigma_1(\beta), \sigma_2(\beta) \leq \varepsilon_D N(\alpha)^{1/2}, \quad \text{Tr}(\beta) = O_D(N(\alpha)^{1/2}).$$

Therefore Proposition 4.5 gives

$$V_\beta^{\text{tr}} = O_D\left(\frac{N(\alpha)^{1/4}}{\sqrt{D}}\right),$$

while every trace row interval has length

$$O(U_\beta) = O_D(N(\alpha)^{1/4}).$$

Hence

$$\Sigma_{\text{row}}(\beta) = O_D\left(\frac{N(\alpha)^{1/4} \log(2 + N(\alpha))}{\sqrt{D}}\right),$$

which is of lower order than $O_D(N(\alpha)/D)$. Substituting these estimates into Theorem 5.1 yields the claimed time bound. $\square$

Proposition 5.7. Assume that $0 \neq \alpha \in O_D$ is representable, and let $\beta = u^{2k} \alpha$ be the balanced element used by Algorithm 3, where $u \in O_D$ is a totally positive unit. Suppose that, while building $S(\beta)$, one stores one root $x_s \in B(\beta)$ with $x_s^2 = s$ for each $s \in S(\beta)$, and that while building $T_2(\beta)$ one stores one witness pair (s_1, s_2) for each $t = s_1 + s_2 \in T_2(\beta)$. Then one can recover roots $y_1, \dots, y_{t_D(\alpha)} \in O_D$ such that

$$\alpha = y_1^2 + \dots + y_{t_D(\alpha)}^2.$$

For lengths 1, 2, 3, 4, the extra work after constructing $S(\beta)$ and $T_2(\beta)$ is linear in $M(\beta) + |T_2(\beta)|$. For length 5, a recursive search over $s \in S(\beta)$ together with four-square reconstruction for $\beta - s$ yields a minimal decomposition with worst-case extra cost $O(M(\beta)^3)$.

PROOF. Let $\gamma = u^{-k} \in O_D$, so that $\alpha = \gamma^2 \beta$. Any decomposition $\beta = z_1^2 + \dots + z_m^2$ therefore rescales to $\alpha = (\gamma z_1)^2 + \dots + (\gamma z_m)^2$.

If $\ell_D(\beta) = 1$, then $\beta \in S(\beta)$ and the stored root x_β gives the desired one-square decomposition. If $\ell_D(\beta) = 2$, then the successful test in the second loop of Algorithm 3 yields $\beta = s + (\beta - s)$ with both summands in $S(\beta)$, and the stored roots of these two squares recover a two-square decomposition. If $\ell_D(\beta) = 3$, then for some $s \in S(\beta)$ the difference $\beta - s$ lies in $T_2(\beta)$; the stored witness pair for $\beta - s$ and the stored root for s recover three roots. If $\ell_D(\beta) = 4$, then the successful test $\beta - t \in T_2(\beta)$ provides two witness pairs in $T_2(\beta)$ and hence four roots. In all of these cases the rescaling by γ gives a minimal decomposition of α .

If $\ell_D(\beta) = 5$, then there exists some $s \in S(\beta)$ such that $\ell_D(\beta - s) = 4$. Trying all $s \in S(\beta)$ and applying the previous four-square reconstruction to each representable remainder eventually succeeds and yields five roots of β ; rescaling again gives five roots of α . The worst-case extra cost is $O(M(\beta) |T_2(\beta)|)$, which is contained in $O(M(\beta)^3)$. $\square$

6 Pythagoras numbers and integral Pythagoras elements

For maximal real quadratic orders, the value of the Pythagoras number is already known by Theorem 2.5. The algorithmic task is therefore to turn that classification into an explicit routine producing both the value $P(O_D)$ and a witness of maximal length.

We first record convenient candidate elements. Their displayed decompositions prove that they are representable by $P(O_D)$ squares or fewer; the extremality will be verified algorithmically by Algorithm 4.

Proposition 6.1. *For the following choices of $a_D \in O_D$, the displayed identity is a decomposition into at most $P(O_D)$ squares:*

$$\begin{aligned} 6 + 2\sqrt{2} &= 1^2 + (\sqrt{2})^2 + (1 + \sqrt{2})^2, \\ 9 + 4\sqrt{3} &= 1^2 + 1^2 + (2 + \sqrt{3})^2, \\ \frac{7 + \sqrt{5}}{2} &= 1^2 + 1^2 + \omega_5^2, \\ 10 + 2\sqrt{6} &= 1^2 + 1^2 + 1^2 + (1 + \sqrt{6})^2, \\ 11 + 2\sqrt{7} &= 1^2 + 1^2 + 1^2 + (1 + \sqrt{7})^2, \\ 12 + 2\sqrt{13} &= 1^2 + 1^2 + 1^2 + \omega_{13}^2 + (1 + \omega_{13})^2. \end{aligned}$$

Moreover, for the generic 5-square cases one has

$$\begin{aligned} 7 + (1 + \sqrt{D})^2 &= 2^2 + 1^2 + 1^2 + 1^2 + (1 + \sqrt{D})^2 & (D \equiv 2, 3 \pmod{4}), \\ 7 + \omega_D^2 &= 2^2 + 1^2 + 1^2 + 1^2 + \omega_D^2 & (D \equiv 1 \pmod{4}). \end{aligned}$$

PROOF. This is a direct verification. $\square$

Algorithm 4: Pythagoras number and integral Pythagoras element

Input : A squarefree integer $D \geq 2$.
Output : A pair $(P(O_D), a_D)$ with $\ell_D(a_D) = P(O_D)$.

```

1 Determine  $p = P(O_D)$  using Theorem 2.5;
2 if  $D = 2$  then
3   | Set  $C \leftarrow \{6 + 2\sqrt{2}\}$ ;
4 else
5   | if  $D = 3$  then
6     | Set  $C \leftarrow \{9 + 4\sqrt{3}\}$ ;
7   | else
8     | if  $D = 5$  then
9       | Set  $C \leftarrow \left\{\frac{7 + \sqrt{5}}{2}\right\}$ ;
10    | else
11      | if  $D = 6$  then
12        | Set  $C \leftarrow \{10 + 2\sqrt{6}\}$ ;
13      | else
14        | if  $D = 7$  then
15          | Set  $C \leftarrow \{11 + 2\sqrt{7}\}$ ;
16        | else
17          | if  $D = 13$  then
18            | Set  $C \leftarrow \{12 + 2\sqrt{13}\}$ ;
19          | else
20            | if  $D \equiv 1 \pmod{4}$  then
21              | Set  $C \leftarrow \{7 + \omega_D^2\}$ ;
22            | else
23              | Set  $C \leftarrow \{7 + (1 + \sqrt{D})^2\}$ ;
24 foreach  $a \in C$  do
25   | if Algorithm 3 applied to  $(D, a)$  returns  $p$  then
26     | return  $(p, a)$ ;
27 return FAIL;
```

Theorem 6.2. Algorithm 4 is correct.

PROOF. The value p computed in the first line is correct by Theorem 2.5. By Proposition 6.1, every candidate in the list C is representable by at most p squares. The classical classification of quadratic Pythagoras numbers, in the form recorded by Krašenský, Raška and Sgallová [13, Theorem 3.1], shows that in each case the chosen candidate is in fact extremal, i.e. has length p . Algorithm 3 is correct by Theorem 4.11, so the verification step returns exactly

when the candidate has length p . Therefore the algorithm outputs a correct pair $(P(O_D), a_D)$, and the FAIL branch is unreachable. $\square$

Remark 6.3. The maximal-order hypothesis matters. For arbitrary quadratic orders there is one further exceptional case, namely the non-maximal order $\mathbb{Z}[\sqrt{5}]$, whose Pythagoras number equals 4 [13, Theorem 3.1]. Extending the present algorithm from O_D to arbitrary quadratic orders should amount to replacing the integral basis, Peters' representability criterion, and the Pythagoras classification by their order-theoretic analogues; the finite-search part of Sections 4 and 5 would remain essentially unchanged.

7 Worked example and computational data

This section gives an end-to-end execution of the algorithm and a small table of computations over $O_{10} = \mathbb{Z}[\sqrt{10}]$.

Example 7.1 (An end-to-end length-5 computation in O_{10}). Let

$$\alpha = 18 + 2\sqrt{10} \in O_{10}.$$

We apply the algorithms step by step.

Step 0: balancing. The Pell-unit balancing step of Algorithm 3 leaves this example unchanged: α is already balanced, so the search is carried out on $\beta = \alpha$.

Representability. Here $D = 10 \equiv 2 \pmod{4}$, $a = 18$, $b = 2$, and

$$N(\alpha) = 18^2 - 10 \cdot 2^2 = 284.$$

Since b is even and the nearest integer to $a/(2D) = 18/20$ is $c_0 = 1$, we get

$$(2Dc_0 - a)^2 = (20 - 18)^2 = 4 \leq 284.$$

Hence Algorithm 1 returns TRUE.

Exact row search. The implementation and the formal Algorithm 2 use only exact trace bounds and sign tests. Here

$$\text{Tr}(\alpha) = 36, \quad V_\alpha^{\text{tr}} = \left\lfloor \sqrt{\frac{\text{Tr}(\alpha)}{2 \cdot 10}} \right\rfloor = 1,$$

so only the rows $v = -1, 0, 1$ can occur. For $v = 0$, Proposition 4.5 gives the enclosure

$$I_0^{\text{tr}}(\alpha) = [-4, 4],$$

and Proposition 4.6 refines this to the exact admissible row $J_0(\alpha) = [-3, 3]$. For $v = 1$, the trace enclosure is $I_1^{\text{tr}}(\alpha) = [-2, 2]$, and the exact admissible row is $J_1(\alpha) = [0, 1]$. For $v = -1$, one again has $I_{-1}^{\text{tr}}(\alpha) = [-2, 2]$, and the exact admissible row is $J_{-1}(\alpha) = [-1, 0]$. Hence

$$B(\alpha) = \{-1 - \sqrt{10}, -\sqrt{10}, -3, -2, -1, 0, 1, 2, 3, \sqrt{10}, 1 + \sqrt{10}\},$$

and squaring these elements yields $S(\alpha) = \{0, 1, 4, 9, 10, 11 + 2\sqrt{10}\}$.

Exact length. One checks immediately that $\alpha \notin S(\alpha)$. Next,

$$T_2(\alpha) = S(\alpha) + S(\alpha)$$

Exact length	Count with $\text{Tr}(\alpha) \leq 40$	Smallest example(s) in $\mathcal{O}_{10}$ at minimal trace
1	11	1
2	22	2
3	14	3
4	9	7
5	2	$18 \pm 2\sqrt{10}$
∞	76	$4 \pm \sqrt{10}$

Table 1. Exact-length distribution in $\mathcal{O}_{10}$ for totally positive elements of trace at most 40.

contains, among others,

$$0, 1, 2, 4, 5, 8, 9, 10, 11, 11 + 2\sqrt{10}, 12 + 2\sqrt{10}, 13, 14, 15 + 2\sqrt{10}, 18,$$

$$19, 20, 20 + 2\sqrt{10}, 21 + 2\sqrt{10}, 22 + 4\sqrt{10},$$

but not $18 + 2\sqrt{10}$. Therefore $\ell_{10}(\alpha) \neq 1, 2$. For the three-square test, the six differences

$$\alpha - s \quad (s \in S(\alpha))$$

are

$$18 + 2\sqrt{10}, 17 + 2\sqrt{10}, 14 + 2\sqrt{10}, 9 + 2\sqrt{10}, 8 + 2\sqrt{10}, 7,$$

and none belongs to $T_2(\alpha)$. Hence $\ell_{10}(\alpha) \neq 3$. For the four-square test, one checks that there is no $t \in T_2(\alpha)$ with $\alpha - t \in T_2(\alpha)$ either. By Theorem 4.9, it follows that

$$\ell_{10}(18 + 2\sqrt{10}) = 5.$$

Since $P(\mathcal{O}_{10}) = 5$ by Theorem 2.5, this is a Pythagoras element of $\mathcal{O}_{10}$.

A small computational run over $\mathcal{O}_{10}$ already illustrates the distribution of exact lengths. More precisely, we considered the finite set

$$\mathcal{E}_{10}(40) = \left\{ \begin{array}{l} a + b\sqrt{10} \in \mathcal{O}_{10} : a, b \in \mathbb{Z}, a + b\sqrt{10} > 0, a - b\sqrt{10} > 0, \\ 1 \leq \text{Tr}(a + b\sqrt{10}) = 2a \leq 40 \end{array} \right\},$$

which contains 134 totally positive elements. The counts were obtained from a direct implementation of Algorithms 1–3; the supplementary script `real_quadratic_sos_table.py` included with the source bundle computes the counts and minimal examples underlying Table 1.

At minimal trace, the elements 1, 2, 3, 7, $18 \pm 2\sqrt{10}$ realise the finite lengths 1, 2, 3, 4, 5, respectively, while $4 \pm \sqrt{10}$ give the first non-representable examples. This small experiment already shows why the element-wise problem is different from the family-wise questions studied in [7, 18]: even within one fixed ring and a narrow trace range, all possible finite lengths and many non-representable elements already appear.

Balancing ablation on a Pell orbit

Let $u = 19 + 6\sqrt{10}$ be the positive norm-one Pell unit of $\mathcal{O}_{10}$ used by the implementation, and set

$$\alpha_k = u^{2k}(18 + 2\sqrt{10}) \quad (k \geq 0).$$

k	$\text{Tr}(\alpha_k)$	$V_{\text{raw}}^{\text{tr}}$	rows_{raw}	$ B $	$ S $	ms_{raw}	$V_{\text{bal}}^{\text{tr}}$	rows_{bal}	ms_{bal}
0	36	1	3	11	6	0.017	1	3	0.017
1	35076	41	83	11	6	1.147	1	3	0.017
2	50579556	1590	3181	11	6	67.798	1	3	0.017
3	72935684676	60388	120777	11	6	4039.467	1	3	0.017

Table 2. Balancing ablation on the Pell orbit of $18 + 2\sqrt{10}$ in $\mathcal{O}_{10}$. Without balancing, the trace-row bound and the number of scanned rows explode along the orbit, while the balanced search remains constant.

By Lemma 2.2(d), one has $\ell_{10}(\alpha_k) = 5$ for every k . Table 2 compares the raw search geometry with the balanced search geometry used by Algorithm 3. The structural columns are machine-independent. The timing columns are medians of three runs of the reference CPython implementation and are included only to illustrate the scale of the balancing effect.

The exact set of relevant roots and squares does not change along the orbit: every balanced search still sees $|B| = 11$ admissible roots and $|S| = 6$ distinct squares. What grows is only the amount of wasted row scanning in the unbalanced coordinates. In particular, the balancing step is not a cosmetic optimisation; it is what keeps the search phase in the norm-sized regime predicted by Corollary 5.6. The timings in Table 2 measure only the search-set construction from Algorithm 2; the current implementation still spends $O(|k|)$ exact preprocessing steps to reach the balanced representative $u^{2k}\alpha$.

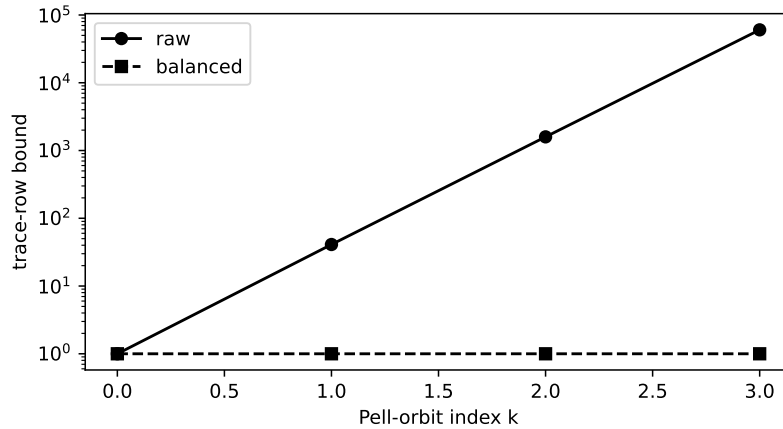


Fig. 1. Raw and balanced trace-row bounds along the Pell orbit in the balancing ablation.

Cross-field distributions and validation

To probe the behaviour beyond $\mathcal{O}_{10}$, Table 3 records exact-length distributions for six representative fields and trace bound 80. The last column $\tau_{\max}$ denotes the smallest trace at which the maximal possible finite length $P(\mathcal{O}_D)$ first appears.

These data illustrate several phenomena already invisible in a single-field table. In the ternary case $D = 5$ every totally positive element of trace at most 80 is representable and the maximal length is already present at trace 7. By contrast, for larger D one sees substantial pools of non-representable elements, together with a delayed first appearance

D	#1	#2	#3	#4	#5	# ∞	$\tau_{\max}$
5	54	576	818	0	0	0	7
10	18	87	112	29	4	270	36
13	35	256	511	78	4	14	24
29	25	130	244	121	22	60	29
53	16	75	116	85	26	126	41
101	14	32	40	28	6	200	65

Table 3. Cross-field exact-length distributions for totally positive nonzero elements of trace at most 80. The column $\tau_{\max}$ records the smallest trace at which the maximal length $P(O_D)$ first occurs.

Fields checked	Trace bound	Elements compared	Mismatches
26	24	1358	0

Table 4. Bounded-trace validation against a separate brute-force coefficient-box implementation.

of maximal-length elements. Even among fields with $P(O_D) = 5$, the distribution of lengths below trace 80 changes considerably with D .

Finally, we compared the package output against a separate brute-force coefficient-box validator. It scans a naive coefficient box, does *not* use the exact row-search engine, Peters’ criterion, or the closed-form Pythagoras classification, and reuses only the exact coefficient arithmetic and embedding-comparison primitives from the ring layer. The sweep covered every squarefree $D \leq 41$ and every nonzero totally positive element of trace at most 24.

This validation sweep does not replace proof, but it provides a useful computational sanity check: it confirms that the exact-length outputs agree with a deliberately naive coefficient-box search on a substantial bounded corpus, without relying on Peters’ criterion, the row-search engine, or the closed-form Pythagoras classification. The sweep concerns lengths only; constructive decompositions are checked separately in the bundled test suite.

Remark 7.2 (Implementation). The companion Python package `quadratic_sos`, available at https://github.com/tomaszkania/quadratic_sos, follows the exact-arithmetic philosophy of the paper all the way through. The representability test already uses only exact coefficient arithmetic by Proposition 3.1. In the search phase, the implementation stores points as coefficient pairs in the basis $1, \omega_D$ and computes each admissible row exactly: for a fixed v , the trace inequality $\text{Tr}(x^2) \leq \text{Tr}(\alpha)$ gives a finite enclosing interval for u , and a directed binary search based on exact sign tests for σ_1 and σ_2 then locates the precise interval of admissible integers on that row. No floating-point approximation is needed for the search. The implementation also applies unit-square balancing before enumeration, using the positive norm-one Pell unit discussed in Remark 5.5. It mirrors the paper’s mathematical assumptions at the API level: the constructor rejects non-squarefree radicands D , and the exact-length and decomposition routines are defined only for nonzero inputs. The repository contains a test suite reproducing the examples and Table 1, together with an executed notebook illustrating the algorithms. The scripts `real_quadratic_sos_table.py`, `balancing_ablation.py`, `cross_field_distributions.py`, and `validation_sweep.py` compute the data underlying the computational study from this section. They are thin wrappers importing the package routines, so the published computations and the library use the same exact-arithmetic engine.

Conclusion

For maximal real quadratic orders, the exact length of a specific sum of integral squares can be computed by a completely explicit finite algorithm. Peters' theorem decides representability, corrected Minkowski bounds control the search geometry, exact trace-row enclosures and directed bisection enumerate the relevant squares without floating-point arithmetic, and a meet-in-the-middle argument distinguishes the exact lengths 1, 2, 3, 4, 5. The same framework also yields an explicit algorithm for the Pythagoras number and for an integral Pythagoras element. The computational study shows in particular that the balancing step is indispensable in practice: along Pell orbits it keeps the search geometry constant while the raw trace-row bound grows explosively.

The present paper is intentionally conservative in its arithmetic input. A natural next step would be to combine the finite-search approach with the continued-fraction theory of indecomposable totally positive integers developed by Dress and Scharlau [4], as in Raška's work on the family-wise problem. Another direction is to import specialised low-rank criteria, for instance Maaß's three-square criterion over O_5 , into the exact-length algorithm as field-specific short-circuits. Finally, it would be natural to extend the algorithm from maximal real quadratic orders to arbitrary quadratic orders, where the exceptional order $\mathbb{Z}[\sqrt{5}]$ should already provide an instructive test case.

Conflict of interest

The author declares that there is no conflict of interest.

Data availability

No external data were gathered or used in this work. All computational tables and timings were generated from the algorithms implemented in the companion repository at https://github.com/tomaszkania/quadratic_sos; the submitted software component is version 0.2.0.

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